Project - High Level Design

on

Enerlytics AI

Course Name: GEN AI

Institution Name: Medicaps University – Datagami Skill Based Course

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MEDICAPS UNIVERSITY

Datagami
"Lead Digital Technology"

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1. Introduction

1.1 Scope of the Document

This High-Level Design (HLD) serves as the primary technical blueprint for northern development of the **Enerlytics AI** platform. It defines the system architecture, component interactions, and technical orchestration required to deliver a specialized energy-domain RAG experience.

1.2 Intended Audience

This document is designed for system architects, lead developers, and technical stakeholders involved in the evaluation and future development of the energy intelligence platform. It provides a granular look at the Generative AI implementation strategies and the infrastructure requirements for scalable document analysis.

1.3 System Overview

Enerlytics AI is an intelligent RAG-driven platform created to assist grid operators and energy specialists in navigating complex technical documentation. Unlike generic AI bots, this system is designed to provide domain-specific insights by leveraging local embedding models and high-performance LLM routing.

The core of the system will be built around a multi-stage validation engine that filters for energy-related keywords before any vector processing occurs. Through the use of vector databases, the bot will retrieve relevant context from technical blueprints and load reports to ensure that AI-generated responses are accurate and grounded. The proposed backend architecture utilizes Python and FastAPI to ensure low-latency communication during streamed analytical queries.

2.Tools and Technologies

1. Frontend Technologies

- **React 18**: Main frontend framework jo reactive aur component-based modern UI banata hai.
- **Vite**: Ultra-fast build tool jo traditional platforms (jaise CRA) ke muqable development time kam karta hai.
- **Vanilla CSS**: Poori application ki styling custom system par based hai taaki maximum flexibility aur original design rahe.
- **Framer Motion**: Smooth scroll animations aur UI components par micro-animations (hover effects) ke liye.
- **Vanta.js (Three.js)**: Website ke piche cinematic 3D interactive energy-neural background banane ke liye.
- **Lucide React**: System icons ka standard set jo UI/UX ko professional banata hai.

2. Backend Technologies

- **Python 3.11**: Application ko lead karne wali main scripting language jo AI tasks ke liye ideal hai.
- **FastAPI**: Ek high-performance asynchronous web framework jo Python ke fast algorithms aur AI models ko lead karta hai.
- **Gunicorn / Uvicorn**: Production-grade ASGI servers jo scaleable backend load handle karte hain.
- **JWT (JSON Web Tokens)**: Secure user sessions aur data isolation manage karne ke liye.

- **Fastapi-mail**: Automated OTP $(\text{One-Time Password})$ aur registration emails bhejne ke liye.

- **PyMuPDF (Fitz)**: PDFs aur DOCX files se accurately text extract karne wala motor engine.

3. AI & RAG Engine

- **Groq (Llama 3.1)**: Duniya ka sabse fast AI inference engine jo massive data se seconds mein jawab generate karta hai.

- **Sentence Transformers (Local)**: Hamara local embedding model (all-MiniLM-L6-v2) jo documents ko machine-understandable vectors mein badalta hai.

- **LangChain**: AI orchestration tool jo document chunking aur RAG flows $(\text{Retrieval-Augmented Generation})$ ko connect karta hai.

- **API Rotator Logic**: Custom-built system jo multiple API keys ko rotate karta hai taaki server kabhi crash na ho.

4. Database & Storage

- **Qdrant Cloud**: Vector database jisme documents ka "Semantic Knowledge" save kiya jata hai fast similarity search ke liye.

- **Redis Cloud**: Session persistent memory aur chat history manage karne ke liye, jo sub-millisecond response time deta hai.

- **Disk Caching**: Chunks aur temporary files ko local storage par process karke clean karne ka automated mechanism.

3. System Design

3.1 Application Design

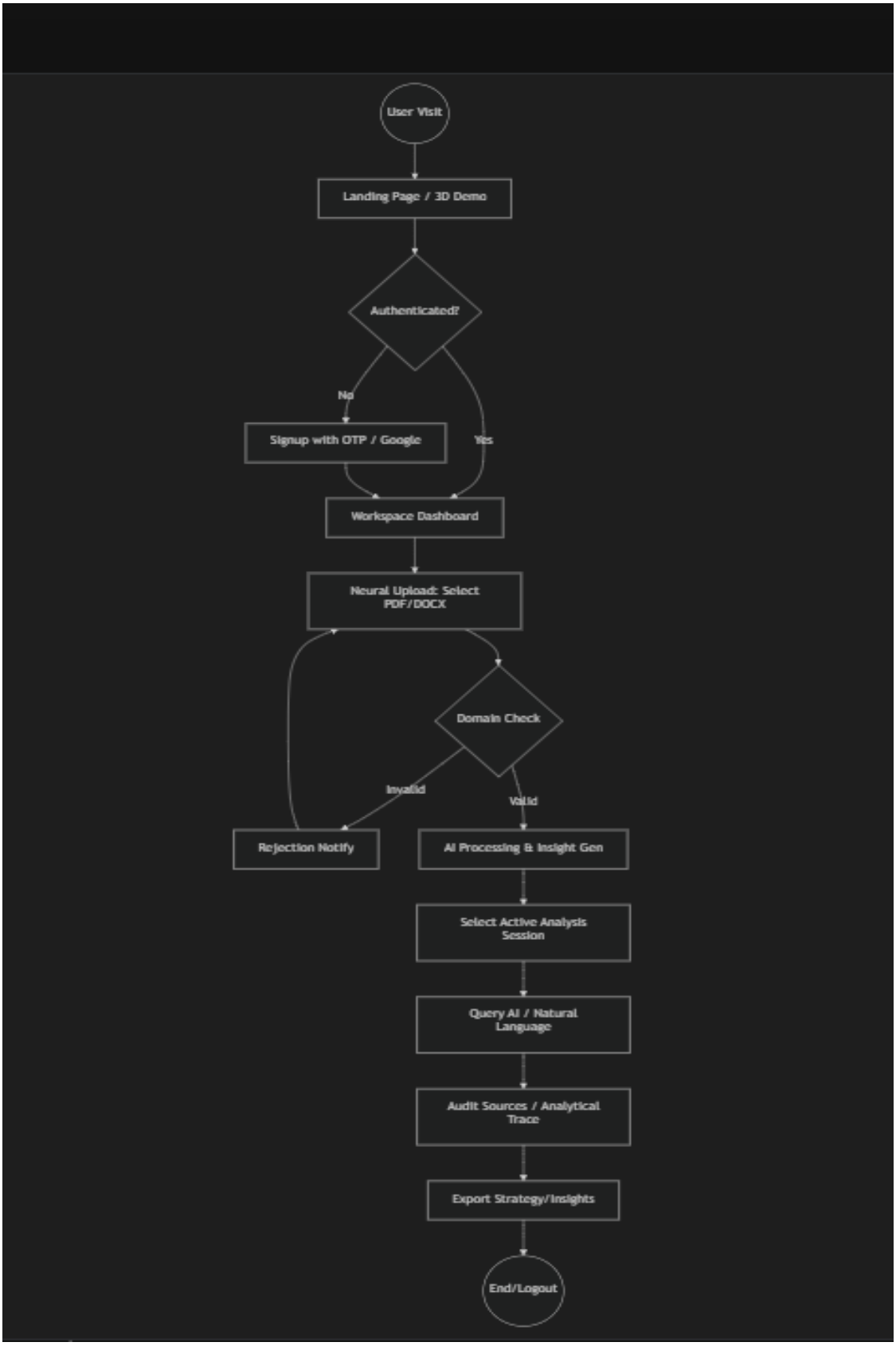
The application is conceived with a modular, layered architecture to facilitate scalability and maintainability. The proposed architecture follows a structured pipeline:

The system will be partitioned into three primary layers:

- **Presentation Layer (React):** Handles the neural-themed user interface, including real-time message streaming and analytical trace visualization.
- **Backend Orchestration (FastAPI):** Serves as the central nerve center responsible for session management, user authentication, and coordinating communication between the AI and storage layers.
- **Intelligence Layer (RAG Pipeline):** Manages the lifecycle of document embeddings, vector retrieval from Qdrant, and response synthesis via the Groq LLM API.

3.2 Process Flow

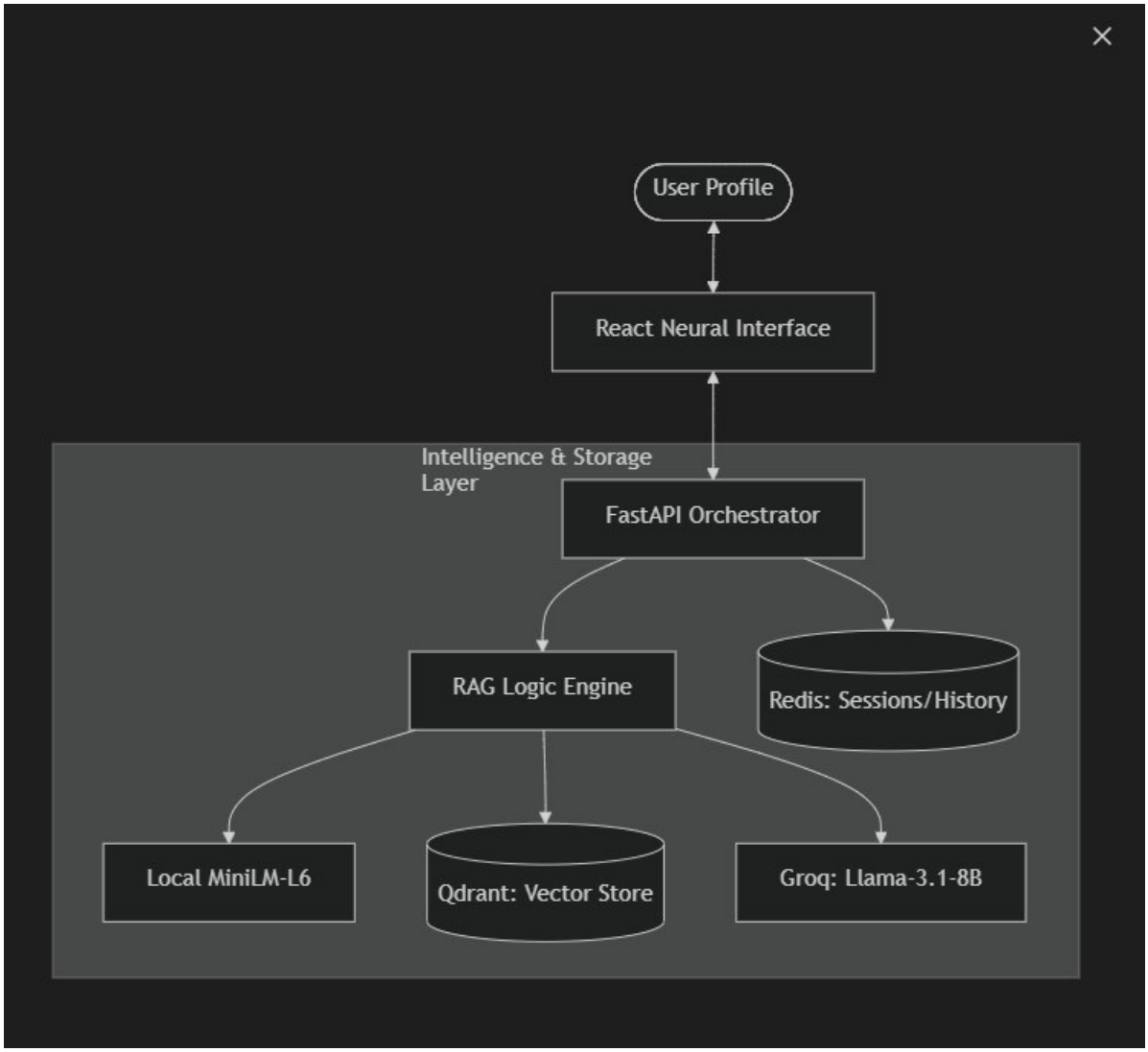
This diagram illustrates how a user navigates and interacts with the platform's features.


User Visit
Landing Page / 3D Demo
Authenticated?
No
Yes
Signup with OTP / Google
Workspace Dashboard
Neural Upload: Select PDF/DOCX
Domain Check
Invalid
Valid
Rejection Notify
AI Processing & Insight Gen
Select Active Analysis Session
Query AI / Natural Language
Audit Sources / Analytical Trace
Export Strategy/Insights
End/Logout

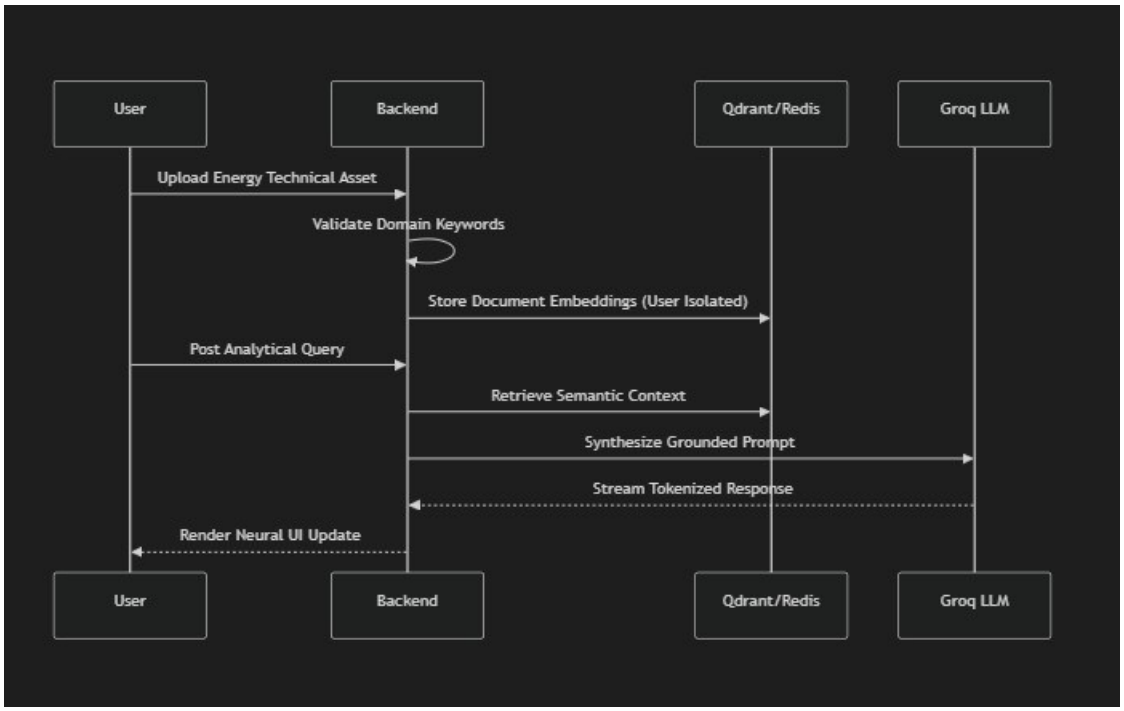
1. **User Visit & Discovery**: The journey begins at the cinematic landing page. A 3D Vanta background and 3D Neural Demo provide a premium first impression, showcasing the AI's capabilities in the energy domain.
2. **Authentication Gateway**: The system performs a state check. Unauthenticated users are routed to a secure signup/login portal featuring Email-based OTP verification for high security or Google OAuth for convenience.
3. **Workspace Dashboard Entry**: Upon successful authentication, a full React state sync occurs, retrieving the user's past documents and analysis sessions from the Redis-backed workspace initializer.
4. **Neural Upload (Data Entry)**: Users drag and drop PDF or DOCX files. This starts a multi-stage ingestion pipeline where files are temporarily cached for processing.
5. **Domain Gating (The AI Guard)**: A critical step where the EnergyAnalysisAgent scans the document content. Using semantic analysis, it determines if the file is genuinely energy-related (e.g., Renewables, Grid, Policy).
    - **Rejection Notify**: If the document is irrelevant, the user is notified immediately to maintain vector database hygiene.
6. **AI Processing & Insight Gen**: Valid documents are chunked and vectorized locally. Simultaneously, the AI generates a "Technical Insight" summary and suggested research topics for the user.
7. **Strategic Analysis (The Chat Bubble)**: Users enter the chat workspace. Using the RAG $(\text{Retrieval-Augmented Generation})$ pipeline, users can ask complex technical questions. The system retrieves document context and historical state to provide accurate, non-hallucinatory answers.
8. **Audit & Verification (Analytical Trace)**: Users can click on a "Trace" button to see exactly which document page or paragraph was used as the source for the AI's response, fostering trust and transparency.
9. **Export & Session End**: Users can export their analysis findings or strategically switch sessions. Logging out triggers a full browser reload to ensure 100% data isolation for the next session.

3.3 Information Flow

Data within the system flows in a structured, uni-directional loop to maintain integrity. User inputs trigger requests to the vectorized storage to find high-similarity "evidence." This evidence is then fused with the user's prompt and passed through a secure LLM pipeline using a "System-Persona" grounding message. The resulting information is processed by the backend to ensure domain grounding before reaching the user interface. Session persistence will be managed via Redis, ensuring that conversation history is contextually preserved during the 10-day session lifecycle.


User Profile
React Neural Interface
Intelligence & Storage Layer
FastAPI Orchestrator
RAG Logic Engine
Redis: Sessions/History
Local MiniLM-L6
Qdrant: Vector Store
Groq: Llama-3.1-8B

Chat Sequence Flow:



3.4 Components Design

- **Neural Upload Module:** Handles multi-threaded document parsing, text normalization, and semantic gating via the EnergyAnalysisAgent.
- **Cognitive Chat Module:** A complex React state manager for handling high-frequency streaming tokens, analytical trace mapping, and markdown rendering.
- **API Key Rotator:** A middleware that ensures 99.9% AI uptime by cycling through multiple Groq keys to mitigate rate limits**.**
- **Vector Sync Service:** Manages the precise alignment between Qdrant vector IDs and the original document page numbers for analytical tracing.

3.5 Key Design Considerations

- **Accuracy & Grounding**: The design prioritizes "Hallucination Protection" by strictly limiting the LLM to the provided energy context.

- **Performance**: The use of local embeddings and a reverse-proxy (Nginx) ensures fast response times in various network conditions.
- **Data Isolation**: The logic ensures that every retrieval operation is strictly filtered by unique user_id metadata.

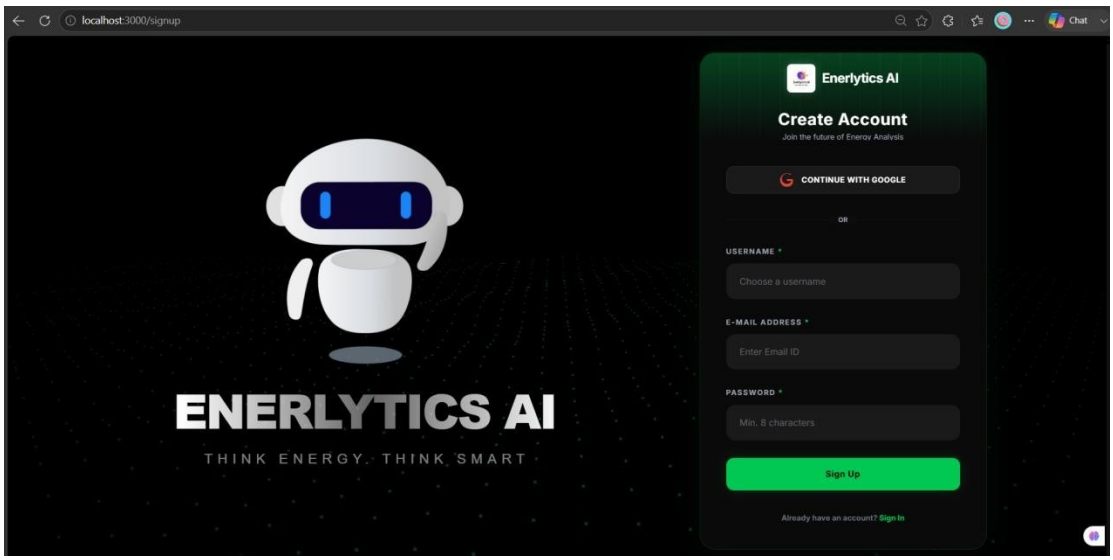
3.6 API Catalogue

The system will expose a set of standardized RESTful APIs including:

- **/auth/signup:** Optimized onboarding and workspace initialization.
- **/api/upload:** Neural document analysis and ingestion.
- **/api/chat:** Multi-modal chat interface with streamed response support.
- **/api/workspace/init:** Unified session recovery and environment synchronization.

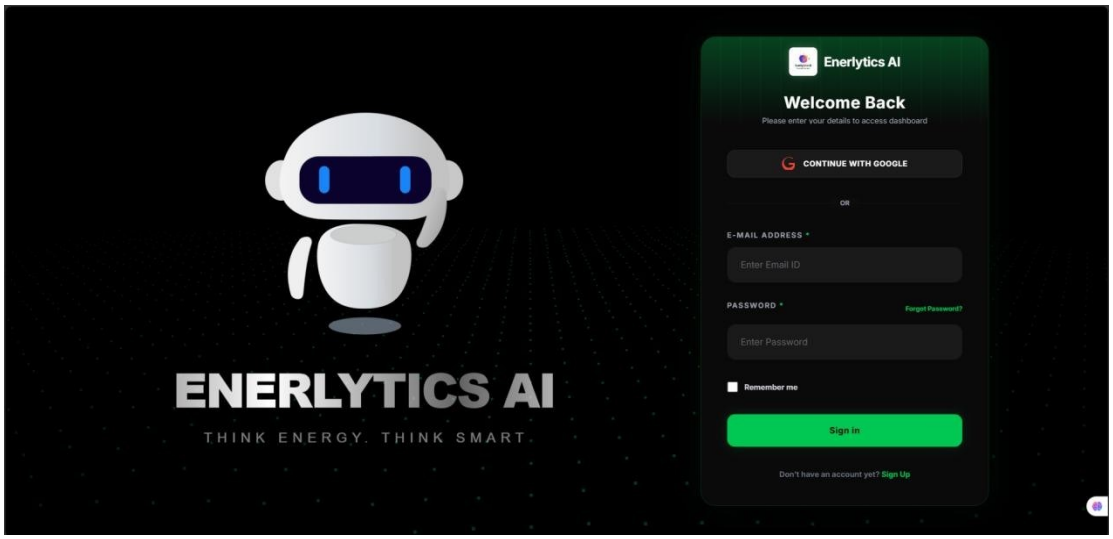
4. Recommended Headings

1. Signup page



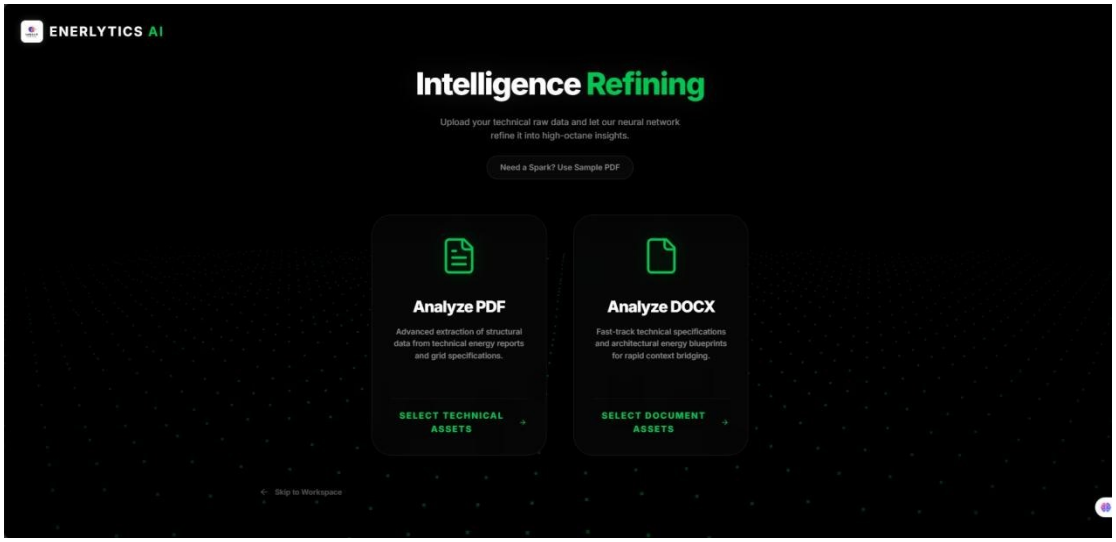
This page allows new users to register for the platform. It asks for basic details like username and email to create an account and get started with the energy analysis tools.

2.Login page



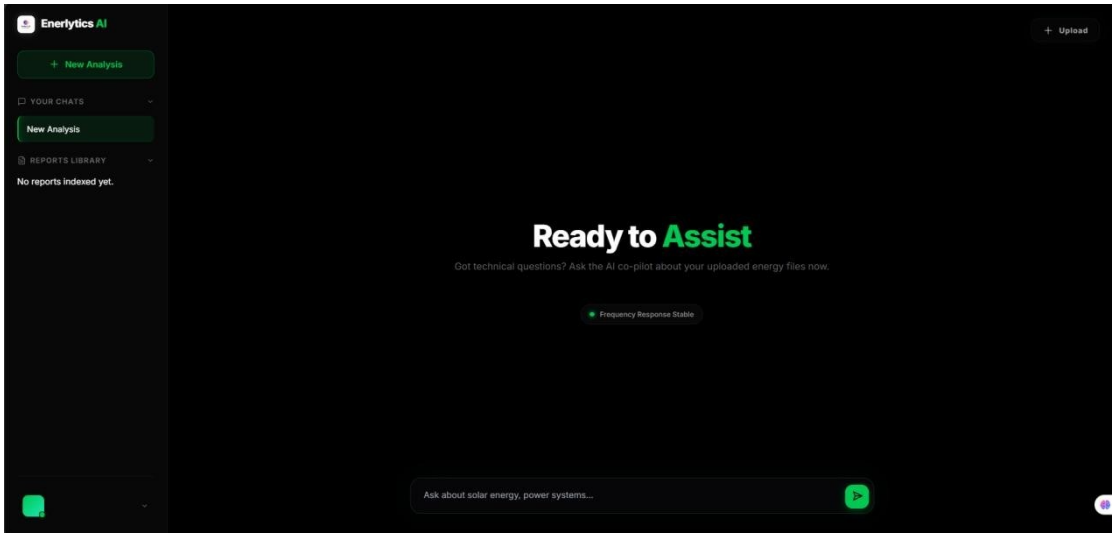
This is the secure login screen where users can sign in using their email or Google account. It provides a quick and professional way for users to access their dashboard.

3. Upload page



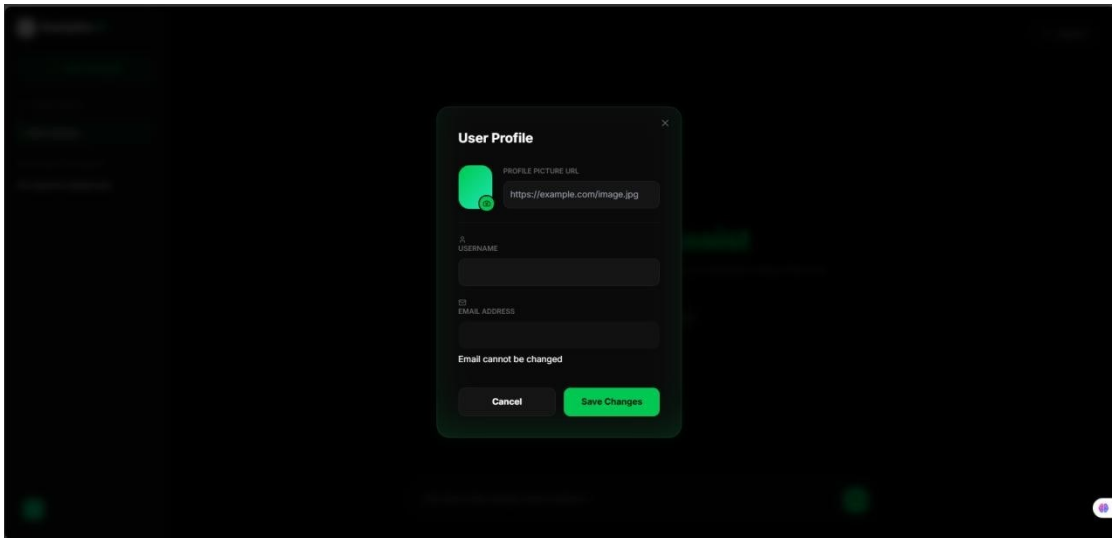
This is the file upload page. Here, users can upload PDF or DOCX documents. Once uploaded, the AI processes these files to extract important technical data and insights.

4.Chat interface



This is the central workspace where users chat with the AI. The sidebar shows saved reports, and the main area allows users to ask questions and get answers based on their uploaded energy documents.

5. Profile model



This is the profile settings window. It lets users update their display name and profile picture URL, while keeping their registered email address fixed for security.

5. References

1. **FastAPI Documentation**: https://fastapi.tiangolo.com/
2. **Qdrant Vector DB Guidance**: https://qdrant.tech/documentation/
3. **Groq Llama-3 API**: https://console.groq.com/docs/
4. **Redis Persistence Layers**: https://redis.io/docs/
5. **Nginx Reverse Proxy Config**: https://nginx.org/en/docs/